ISOMETRIC

Biochar Storage in Soil Environments

Storage Module

This module details durability and monitoring requirements for biochar spreading and storage in soil environments.

v1.2

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1.0

Introduction

The Durability of a Carbon Dioxide Removal (CDR) process refers to the length of time for which carbon dioxide (CO_2) or carbon dioxide equivalents (CO_2e) is removed from the Earth's atmosphere and therefore cannot contribute to further climate change. This Module details the durability, reversal risks and requirements for storage of carbon as biochar in soils. This Module is intended for use in conjunction with other Isometric Protocols and Modules, and assumes the following:

- The full quantification of the net tonnes of carbon dioxide equivalent CO_2e removal for crediting occurs following the Isometric Biochar Production and Storage Protocol.
- All environmental and social safeguards have been followed according to Section 5 in the Biochar Production and Storage Protocol.

The information and requirements outlined within this Module are based on the best available science at the time of writing. This Module will be reviewed at a minimum every two years and/or when there is an update to scientific published literature which would affect net CO_2e removal quantification or the monitoring guidelines outlined in this Module, and/or in line with changes in scientific consensus regarding the durability of biochar in agricultural soils.

1.1

Applicability

This storage Module applies to Projects or processes which apply biochar to productive or amenity land to store captured CO_2. Biochar can be applied as pellets or pieces of biochar, or in combination with additional organic material (e.g. mixed with compost).

Land uses considered under this Protocol include, but are not limited to:

- Agricultural soils - as defined by the United Nations Food and Agriculture Organization -permanent crop land, meadows and pastureland[1].
- Forestry soils - soils within the boundary of a forest or wood under active or inactive forest management.
- Amenity and recreational land - land designated to provide recreational, aesthetic or environmental benefits for example, parks, green spaces and golf courses.

Projects applying biochar in soil storage environments not listed above may be applicable under this Protocol, provided they obtain prior approval from Isometric. Examples of similar eligible environments may

include rangelands, urban landscapes, horticulture, landscaping or other land on which crops are grown.

To be considered eligible Projects must clearly demonstrate in the Project Design Document (PDD) that:

- No land management practice would be undertaken that would pose a risk to storage greater than that associated with agricultural soils.
- All relevant environmental and social safeguards are met, with explicit consideration of local regulations specifically for that environment, in accordance with Section 5 of the Isometric Biochar Production and Storage Protocol.

The following Projects are explicitly ineligible under this Module, include but are not limited to:

- Projects that lead to a sustained, net decrease in crop yields or land productivity.

1.2 Background

This storage Module is associated with the Biochar Production and Storage Protocol, which outlines the requirements and procedures for calculating net CO_2e removal achieved through the production of biochar. This Module specifically addresses the site conditions for biochar storage in soils, including quantification of CO_2e_{stored}. For more information on pyrolysis conditions and biochar characterization, please refer to Section 8.3 of the Biochar Production and Storage Protocol and Appendix I of this Module.

Soils have the potential to act as a significant carbon Sink, as evidenced by the magnitude of soil organic carbon (SOC) stocks - which exceed that of plant matter and atmospheric carbon combined[2]. Utilizing this potential will be important for meeting ambitious climate goals, such as those put forth by the IPCC[3]. SOC stocks are dynamic, influenced by the nature of soil management, local climate and soil type[4]. Natural accumulation of SOC often relies on the input of labile (easily broken down) organic matter, which can result in carbon being re-released to the atmosphere on timescales that are too short for meaningful climate change mitigation[5]. As an alternative, biochar, produced through pyrolysis, offers similar soil health benefits while containing a higher stable carbon fraction. This carbon can persist in soils over much longer timescales[6].

Although there is growing scientific consensus on the stability of biochar in soil environments[7], there are still some key areas where further research is needed. While biochar is more stable than non-pyrolyzed carbon, the mean residence time (MRT) of biochar will depend on its physical and chemical characteristics prior to

application, which is highly influenced by feedstock composition and pyrolysis conditions[8], as well as environmental conditions at the storage location[9].
Higher pyrolysis temperatures are generally associated with the production of more stable biochar[10]. The chemical stability of biochar is considered to be highly related to the formation of aromatic ring structures during pyrolysis. The arrangement and size of these aromatic rings structures, particularly those contributing to the persistent aromatic carbon (PAC) pool, contribute to the stability and resistance to degradation in soil environments. As such, direct and proxy measurements of chemical stability e.g., carbon aromaticity through H/C_{org} ratios[8], as well as comparison with stable organic geological proxies e.g. inertinite[11] have become standard determinants of biochar stability.

As outlined in Section 3, only biochars with a hydrogen-to-organic carbon ratio (H/C_{org}) of <0.5 are considered, which provides high confidence that the biochar is stable on Crediting time horizons of at least 200 years.

Physical degradation of biochar particles, though abiotic, biotic and mechanical turnover can take place over months or years upon application to soil[12]. This degradation primarily affects biochar particle size, and to a lesser extent, specific surface area and porosity,[13]. However, these factors are not necessarily indications of biochar decay, and the carbon content of the biochar can remain stably stored despite this potential breakdown[13,14].

This Module describes how the biochar characteristics described in Section 3 are used to quantify the number of Credits that are issued for a Project applying biochar to soils.

Isometric offers two options for Crediting under this Module:

- 200 year durability based on H/C_{org} ratio and soil temperature based on a conservative interpretation the Woolf et al., (2021)[15], modified for conservatism.
- 1000 year durability based on the random reflectance value of the biochar, compared to inertinite as a proxy for geologically stable carbon, using Sanei et al., (2024)[11]. This method discounts the reactive carbon fraction of the biochar, and therefore only accounts for the recalcitrant fraction. Furthermore this portion is credited at one standard deviation below the mean to ensure a conservative estimate of durability.

It also includes details of the environmental conditions that must be met and documented in the PDD to ensure that biochar carbon is durably stored in surface land applications for at least 200 years.

Currently, there is a lack of commercially available and widely accepted methods to distinguish biochar organic carbon from SOC, once biochar has been incorporated into soil, particularly at small

particle sizes[16]. This is further complicated by the potential for biochar vertical and lateral mobility in the soil profile, through mechanical, abiotic and biotic factors. This makes observing direct changes in SOC stocks, and associating those changes with biochar application very challenging. Spectroscopic techniques such as near- and mid-infrared spectroscopy (NIRS/MIRS), coupled with comprehensive reference databases, show potential for distinguishing carbon from other SOC fractions[16]. However, these databases are currently limited in scope, require further verification, have a high cost, and are not well-suited for routine analysis.

Therefore, quantifying the durability of biochar, based on the current best available science, focuses on rigorous characterization of the biochar produced by pyrolysis in order to determine the fraction that remains stable beyond the desired Crediting time horizon. This should be coupled with conservative treatment of the uncertainty associated with that calculation (see Section 6.5 of the Biochar Production and Storage Protocol). The characterization and quantification requirements of biochar carbon stability in soils will be updated in future versions of this Module, in line with the best available peer-reviewed scientific literature, to ensure the highest standards for carbon credit issuance are maintained.

1.3 Language and Compliance Requirements

This Protocol employs specific terminology to clearly distinguish between mandatory and recommended actions:

Mandatory Requirements:

- "Must" and "required" indicate obligations. It is necessary for Project Proponents to implement these measures without exception.

Recommendations:

- "Recommended", "may" and "should" indicate best practices that are strongly encouraged but not mandated. While compliance with these items is not required, suppliers are expected to provide justification when choosing not to implement recommended measures.

All Project Proponents must acknowledge understanding of this distinction and confirm their ability to meet all mandatory requirements before submitting their PDD.

2.0

Environmental and Social Safeguards

2.1

Safeguarding of Storage Sites

Productive and healthy soils are one of most important, non-renewable resources on the planet, providing humanity with food (98.8% of daily calorie intake[17]), as well as a significant array of additional provisioning, regulating, cultural, and supporting ecosystem services[18]. Thus, maintaining land productivity and health is critical not only for the environmental and social sustainability of CDR project, but also society more broadly. In line with the Isometric standard (Section 3.7), The Project Proponent must ensure that The Project, at minimum, should not result in net environmental or social harm, and comply with all regulations within the jurisdiction (Section 3.6).

Application of biochar should be at or below rates specified in regulation in the jurisdiction where application is taking place, and taking into account soil type and current management regime. On application, precautions should be taken to minimize airborne dust and, where appropriate, biochar should be incorporated into soil immediately, in order to avoid visible black layers and lowered soil albedo.

As shown in **Table 2**, Isometric requires Project Proponents to measure several parameters that are critical for mitigating potential negative impacts on agricultural productivity. These include total nitrogen, pH, salt content, water-holding capacity, and nutrient concentrations (P, K, Mg, Ca, Fe).

If productivity or soil quality are demonstrated to be adversely affected, The Project Proponent must:

- Collaborate with land managers or owners to implement soil management practices that maintain or enhance soil quality. Examples of such practices include use of organic fertilizers, or diversifying crop rotation and utilizing cover crops.
- Provide technical support, training and resources to help land managers adapt to any changes in soil conditions due to the CDR project.

Additionally:

- If pre-existing heavy metal concentrations in soils exceed applicable regulatory limits or guidance, The Project may still be eligible for crediting under this Protocol. However, The Project Proponent must provide evidence of the existing elevated concentrations and implement specific remediation strategies to address the contamination or render the elements inaccessible. These strategies could include altering the variety or quantity of

biochar applied, implementing soil amendments or introducing phytoremediation practices using plants adept at absorbing heavy metals. These practices should only be undertaken where they do not contradict applicability criteria in the relevant storage Module. Any Project with pre-existing elevated heavy metal concentrations which further increases soil contamination unmitigated, will not meet the safeguarding criteria of this Protocol.

In agricultural settings, crop yields may be reported on an annual basis. Crop yield for the project area can be evidenced using historical data from farms directly, farming cooperatives and/or public databases. In the case that a sustained (> 3 years) net crop decrease is reported then Isometric may request additional soil analysis to be conducted consistent with requirements in Section 5.2.

2.2 Co-Benefits

Biochar application to soil may have an impact on properties such as pH, porosity, cation exchange capacity (CEC) and nutrient retention properties (related to surface charge characteristics)[19][20]. However, when produced in accordance with the Biochar Production and Storage Protocol and meeting the chemical and physical standards outlined in Section 3, high-quality biochar applied at appropriate rates is not expected to negatively affect soil health or productivity. In addition to carbon sequestration potential, the application of biochar to soils may deliver a range of co-benefits, including, but not limited to, the following:

- Improving the buffering capacity of soil pH[21]
- Decreased bulk density which may reduce soil compaction[22]
- Increased soil water holding capacity[23]
- Decreased bioavailability of heavy metals[24]
- Increased crop productivity and quality[25]
- Increased nutrient retention capacity[26]

Specific co-benefits are likely to vary based on the specific biochar characteristics and soil conditions and type[27].

In particular, the crop rotation, reported crop yields, soil quality (e.g. pH, nutrient and moisture contents) and fertilizer use may be reported in the PDD to demonstrate evidence of co-benefits.

3.0

Biochar Characterization

3.1

Overview

This Section provides the requirements for the characterization of biochar for durable storage in soil environments, to determine if the material is eligible for Crediting under the Biochar Production and Storage Protocol. Durability refers to the length of time for which CO_2 is removed from the Earth's atmosphere. The durability of biochar will depend on its physical and chemical characteristics as well as the storage site conditions[9,20,28]. Biochar physical and chemical characteristics will be highly influenced by the biomass feedstock type and pyrolysis conditions. This section will not set requirements or guidelines for biomass feedstock eligibility or pyrolysis conditions. Please refer to the Biomass Feedstock Accounting Module and Section 9 of the Biochar Production and Storage Protocol for guidance and discussion on these two topics.

Some of the required measurements in this Section have minimum or maximum thresholds that determine eligibility for Crediting by Isometric. Others may be required but have no associated eligibility threshold. Additionally, certain measurements are not mandatory; however, Project Proponents are strongly encouraged to measure and report them to support scientific progress in understanding biochar durability in soil. Analytical methods provided are examples of eligible methodologies, but they are not the only ones permitted.

For each parameter, the selected methodology or analytical technique, along with an appropriate standard reference (e.g., ISO, ASTM, DIN), where applicable, must be specified in the PDD.

All laboratories used for analysis must conform to ISO 17025 or equivalent. Alternatively, laboratories may be eligible in consultation with Isometric if they can provide adequate QA/QC data. Sample preparation should be performed in adherence to ISO 13909-4:2025.

Biochar must be characterized prior to soil application to ensure environmental safety and suitability for CO_2 removal.

3.2

Physical Characteristics

This Section specifies the recommended physical analyses for biochar characterization, for the purpose of supporting data collection given the current early stage of biochar durability quantification. Physical properties of biochar may affect the degradation of biochar in soil, however there is not yet any evidence that the physical properties of biochar would materially affect its durability of carbon on the crediting time horizon. While a significant body of research exists on biochar application to soil, uncertainty regarding the long-term impact of

biochar on soil still exists. As such, the following analyses of biochar's physi Add comment are recommended; however, no specific eligibility thresholds will be applied.

Table 1: Recommended Measurements of Biochar physical properties

PROPERTY	EXPECTED UNIT	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Specific surface area	$m^2 g^{-1}$	–	[Brunauer-Emmett-Teller (BET)] method ISO 9277:2022	Surface area of applied material may influence a number of biochar stability and soil health characteristics, including: SOC stocks, adsorption rates, water retention and porosity	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended
Porosity	%	–	Mercury porosimetry and gas adsorption ISO 15901-2:2022	Porosity is an indicator of water adsorption potential	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended
Particle size distribution	% by size fraction	–	Sieving ISO 565:1990 (particle sizes between 125 mm and 45 µm) or laser diffraction ISO 13320:2020	Particle size distribution gives an estimate of the range and proportion of different sized particles within a biochar sample. Generally, larger biochar particles will degrade more slowly.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended

3.3 Chemical Characteristics

Project Proponents must use the following analyses of the chemical composition of biochar to assess the reactivity potential of biochar-associated carbon in the soil storage environment.

Some of these measurements will be used in the quantification of CO_2e_{stored}, as outlined in Section 5. The required and recommended measurements listed below investigate multiple mechanisms of reactivity (or prevention of), including aromaticity and aromatic condensation, functional groups, and volatility. The redundancy of characterizing reactivity potential via different mechanisms serves to reduce the uncertainty surrounding the durability of biochar, and provides multiple indicators of confidence that durability will exceed the crediting time horizon. All analyses documented in the table below meet or exceed the WBC standards.

N.B. All results pertaining to the calculation of carbon removal must be reported on a dry basis. Reporting on a dry basis provides a standardized, stable reference point for comparing material properties, making the data repeatable and reliable regardless of the sample's water content at the time of testing.

Table 2: Recommended and required measurements of biochar chemical properties

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
$Total\ Carbon\ Content$	% (weight/weight)	–	ISO 29541:2025 or ISO 16948:2015 or ASTM D5373-21	The carbon content of applied biochar is necessary for the calculation of C_{org} and thus CO_2e_{stored}, in accordance with Section 8.3 of the Biochar Production and Storage Protocol. See Section 8.3.1 of the Biochar Production and Storage Protocol for carbon content sampling guidance.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Moisture Content	% (weight/weight)	–	ISO 18134-1 or ASTM D1762-84	The moisture content of applied biochar is necessary for the quantification of CO_2estored, in accordance with Section 8.3 of the Biochar Production and Storage Protocol. See Section 8.3.1 of the Biochar Production and Storage Protocol for carbon content sampling guidance.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Inorganic Carbon Content (C_{inorg})	% (weight/weight)	-	ISO 16948:2015 or ASTM D4373-02 or DIN 51726: 2004-06	Measurement of C_{inorg} in biochar is required to accurately differentiate organic carbon (C_{org}) from $Total\ Carbon\ Content$, which may include both inorganic and organic forms. Only organic carbon is credited for under this Protocol and Module.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
					number of 3 samples per sampling.	
Total Hydrogen (H)	% (weight/weight)	-	ISO 29541:2025 or ISO 16948:2015 or ASTM D5373-21	Measurement of H is required to calculate the H/C_{org} ratio.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Total Nitrogen (N)	% (weight/weight)	-	[ISO 29541:2025] or ISO 16948:2015 (https://www.iso.org/standard/86983.html) or ASTM D5373-21	Nitrogen is a key component that influences biochar's properties and its potential applications, including its use as a soil amendment.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	
Total Oxygen (O)	% (weight/weight)	-	ISO 16948:2015 or DIN 51733:2016-04 or by difference (sum of % carbon hydrogen, sulfur and ash subtracted from 100)	Measurement of total O is required to calculate the O/C_{org} ratio.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
Total Sulfur (S)	% (weight/weight)	-	ISO 15178:2000 or ISO 16994:2015 or DIN 51724-3:2012-07	Sulfur is a key component that influences biochar's properties and its potential applications, including its use as a soil amendment.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Required
Organic Carbon (C_{org}) Content	% (weight/weight)	-	Calculation	C_{org} is derived from the $Total\ Carbon\ Content$ minus the inorganic carbon content in the sample. C_{org} represents the initial total of organic carbon stored in biochar. This is the basis on which CO_2e_{stored} is calculated taking into account the mass of biochar applied and the durability of the carbon.	CMeasure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Molar H/C_{org} ratio	Ratio	< 0.5	Calculation	Molar H/C_{org} is derived from the H and C_{org}, calculated % values are divided by their respective atomic weight. Low H/C_{org} ratios indicate the presence of significant amounts of aromatic compounds within the biochar, which are highly stable and conducive to long-term stability of sequestered biochar in soil. For the 200 year crediting option, this is used to model biochar durability.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
Molar O/C_{org} ratio	Ratio	< 0.2	Calculation	Molar O/C_{org} is derived from the O and C_{org}, calculated % values are divided by their respective atomic weight. The O/C_{org} ratio indicates the presence of functional groups, with lower ratios indicative of fewer functional groups. A lower abundance of functional groups is favorable for biochar permanence, as these groups can serve as reactive sites on the biochar surface and potentially enhance degradation processes. C-O bonds are more labile than C-C bonds. Furthermore, the O/C_{org} ratio is required to verify that low H/C_{org} ratios genuinely reflect a high degree of aromaticity, rather than the presence of oxygenated aliphatic carbon.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Ash Content	% (weight/weight)	–	ISO 18122 or ISO 1171 or DIN 51719	Measurement of ash content in biochar is important because it represents the inorganic, non-combustible fraction remaining after complete combustion. Ash content can influence soil pH, nutrient availability, and biochar's capacity to retain water and nutrients when applied to soil.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required
Bulk Density (< 3 mm particle size)	kg m^{-3}	–	ISO 17828: 2025 or VDLUFA-method A 13.2.1	This measurement standardizes particle size to < 3 mm to provide a	Minimum number of 3 samples.	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
				consistent metric for comparing different biochar samples. It is primarily used for research and characterization purposes, as it eliminates the variability caused by particle size distribution. Bulk density of the < 3 mm fraction also provides insights into the porosity and compaction characteristics of the finer material.	Measured at project validation unless feedstock, reactor or process parameters change.	
Volatile matter content (VMC)/ Volatile Compounds	% (weight/weight)	-	ASTM D1762-84 or DIN 51720: 2001-03	VMC is indicative of the level of carbonisation, stability, and reactivity of biochar. A higher VMC suggests greater reactivity, while a lower VMC means reduced interaction with soil components.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Recommen
pH	-	-	ISO 10390:2021	Biochar pH reflects its potential impact on soil health, quality, and microbial activity when used as a soil amendment. Measuring pH will assess biochar's influence on these factors. Additionally, pH may indirectly affects biochar durability. However, there is no specific eligibility threshold for biochar pH.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Required
Salt content	g kg^{-1}	-	ISO 10390:2021	Salt content is an important parameter in biochar characterization because elevated levels of soluble salts can negatively affect soil health and plant growth	Minimum number of 3 samples. Measured at project validation unless	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
				when the biochar is applied.	feedstock, reactor or process parameters change.	
Water holding capacity (WHC)	%	-	ISO 14238, annex A	Water holding capacity (WHC) is an important property of biochar because it influences soil moisture retention, plant-available water, and overall soil structure when the biochar is applied.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Recommen
Declaration of the nutrient content (P, K, Mg, Ca, Fe)	g kg^{-1}	-	[DIN EN ISO 11885:2009-09] Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) analysis applied following an appropriate digestion step. (https://www.dinmedia.de/en/standard/din-en-iso-11885/118931490)	Declaration of the nutrient content of biochar is important because these elements (phosphorus, potassium, magnesium, calcium, and iron) contribute to soil fertility and plant nutrition when biochar is applied.	Minimum number of 3 samples. Measured at project validation unless feedstock, reactor or process parameters change.	Required
Heavy metals (lead (Pb), cadmium (Cd), copper (Cu), nickel (Ni), mercury (Hg), zinc (Zn), chromium (Cr), and arsenic (As)	mg kg^{-1} or g t^{-1} DM (directly equivalent)	Pb = 300 g t^{-1} DM, Cd = 5 g t^{-1} DM, Cu = 200 g t^{-1} DM, Ni = 100 g t^{-1} DM, Hg = 2 g t^{-1} DM, Zn = 1000 g t^{-1} DM, Cr = 200 g t^{-1} DM, As = 20 g t^{-1} DM	ISO 16967:2015 or ISO 17294-2:2023 or ISO 16968:2015 or ISO 23380:2022	Quantification of heavy metals in biochar is essential to ensure environmental and human health safety. Elevated concentrations of metals such as lead (Pb), cadmium (Cd), copper (Cu), nickel (Ni), mercury (Hg), zinc (Zn), chromium (Cr), and arsenic (As) can pose risks of soil and water contamination, bioaccumulation in crops, and potential toxicity to soil organisms. Measuring and declaring heavy metal content	Minimum number of 1 sample(s), representative of the production process, measured at project validation unless feedstock, reactor or process parameters change.	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
				allows for verification against regulatory limits and safeguards the suitability of biochar for soil application.		
Polycyclic aromatic hydrocarbons (PAHs)* - U.S. Environmental Protection Agency (EPA) 16 and European Food Safety Authority (EFSA) 8	mg kg^{-1} or g t^{-1} DM (directly equivalent)	EPA 16 = declaration, EFSA 8 = 1 g t^{-1} DM	Calculated from DIN EN 17503 or EPA 8270E with preparation method: EPA 3546	Measurement of PAHs in biochar is required to assess potential environmental and human health risks. PAHs are a group of organic contaminants that can form during pyrolysis, and some are known to be carcinogenic or otherwise toxic. The EPA 16 set refers to the 16 priority PAHs identified by the U.S. EPA, while the EFSA 8 subset refers to the eight PAHs prioritized by the EFSA for food and feed safety. Quantifying these compounds ensures that biochar complies with international safety standards and is suitable for soil application.	Minimum number of 1 sample(s), representative of the production process, measured at project validation unless feedstock, reactor or process parameters change.	Required
Polychlorinated dibenzodioxins/-furans (17 PCDD/F)	ng kg^{-1} DM	PCDD/F: 20 ng kg^{-1} DM	DIN EN 16190 or EPA Method 8290A	Measurement of the 17 toxicologically relevant polychlorinated dibenzodioxins and dibenzofurans (PCDD/F) is required because these persistent organic pollutants can form as [by-products] during the pyrolysis of certain feedstocks. PCDD/F compounds are highly toxic, bioaccumulative, and can pose significant risks to human health and the environment. Quantifying their levels ensures that biochar complies with	Minimum number of 1 sample(s), representative of the production process, measured at project validation unless feedstock, reactor or process parameters change.	Required

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENI OR REQUIRE
				international safety limits and is suitable for soil application without introducing harmful contaminants.		
Polychlorinated biphenyl (12 WHO PCB)	mg kg^{-1} DM, sometimes reported in µg kg^{-1} DM (to convert divide by 1000)	PCB: 0.2 mg kg^{-1} DM	DIN EN 16167 or Analytical Method: EPA 8082A with preparation method: EPA 3546	Measurement of the 12 dioxin-like polychlorinated biphenyls (WHO-PCBs) is required because these compounds are toxic, persistent, and can bioaccumulate in the environment. They may be introduced through contaminated feedstocks or form as trace by-products under certain production conditions. Quantifying WHO-PCBs ensures that biochar meets international safety standards, safeguarding soil health, food chains, and human health when applied to land.	Minimum number of 1 sample(s), representative of the production process, measured at project validation unless feedstock, reactor or process parameters change.	Required
Bulk Carbon Bonding State	% by carbon bonding type (aromatic, aliphatic, carbonyl)	-	NMR spectroscopy	High aromaticity and aromatic condensation are shown to increase MRT by an order of magnitude. High degrees of aromatic condensation result in biochar that is less prone to microbial activity.	Minimum number of 1 sample(s), representative of the production process, measured at project validation unless feedstock, reactor or process parameters change.	Recommen
External surface carbon bonding state composition	Relative proportion (%) of each functional group out of	-	X-ray photoelectron spectroscopy (XPS)	Biochar degrades from the outside in. If the exterior of the biochar particles has a different chemical than the center,	Minimum number of 1 sample(s), representative of the	Recommen

PROPERTY	EXPECTED UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRE
	the total surface carbon detected			that affects degradation rate. Comparing external to internal composition without depth profiling can be done by comparing XPS of in-tact particles to Raman/NMR of pulverised samples OR XPS of pulverised and unpulverised samples. In either case, the sample preparation should be specified in the PDD. Pulverising samples ensures the average chemical composition throughout the particle is measured, whereas the surface composition of in-tact particles can be characterised by XPS.	production process, measured at project validation unless feedstock, reactor or process parameters change.	

*A note on PAH requirement - PAH is required as outlined in Table 2, unless it can be demonstrated that stringent risk mitigation has been carried out, this is applicable to high tech, continuous production processes only. This would include pre-agreeing the risk mitigation with Isometric and detailing this in the PDD. Risk mitigation may include the following:

- Demonstrating a sufficiently high pyrolysis temperature to ensure thermal cracking of PAHs.
- Reactor design choices such as:
  - Increased reactor residence time
  - Testing to show a thermal destruct unit removes PAHs to negligible levels during pyrolysis, coupled with evidence that those operating conditions for pyrolysis are maintained to remove the need for testing of PAHs.
- Evidence of how post-pyrolysis treatment of biochar removes PAHs.
- Previous evidence of the same production process producing acceptably low PAH concentrations.

Table 3: Additional chemical characterisation required for 1000 year durability

PROPERTY	UNIT	THRESHOLD	RECOMMENDED ANALYTICAL METHODOLOGY	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Random Reflectance (R_0)	%	> 2% for inertinite (creditable fraction)	ISO 7404-5:2009, minimum 500 individual measurements	Random reflectance is an indicator of aromaticity, aromatic ring unit size and condensation. A R0 value greater than 2% has been proposed as a benchmark for quantifying the permanent pool of carbon in a biochar [11]. The R0 frequency distribution histogram is used to decide what fraction of biochar above this benchmark can be classified as geologically inert [11].	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required only for 1,000 year durability crediting
Reactive Organic Carbon and Residual Organic Carbon ($C_{non-reactive}$)	%	-	Thermogravimetric analysis e.g., Hawk, Rock-Eval® or equivalent. The sample is subjected to re-pyrolysis using a standardized heating procedure: it is first held isothermally at 300 °C, then heated at a rate of 25 °C per minute until reaching 650 °C. During this stage, the reactive organic carbon is volatilized and quantified. The remaining material, referred to as “residual organic carbon,” is subsequently measured by combustion at temperatures up to 850 °C.	Measurement of reactive organic carbon in biochar is important because this fraction represents the more labile, easily degradable component of organic carbon. Elevated levels of ROC can reduce biochar’s long-term carbon stability, as it is more susceptible to microbial decomposition and mineralization in soil. Random reflectance values are subsequently only applied to the residual, stable fraction of biochar.	Measure every production batch as per Method A or B applicable, as defined in Section 8.3.2 of the Biochar Production and Storage Protocol. Minimum number of 3 samples per sampling.	Required only for 1,000 year durability crediting

4.0

Sampling Guidance, Laboratory Requirements, Data Quality

4.1

Sampling Guidance

For the required measurements in **Table 2**, samples should be taken using the same sampling regimes outlined in Section 8.3.1 of the Biochar Production and Storage Protocol for measuring carbon content.

A batch associated with any one project may have a unique history or set of characteristics that could require individual consideration for recommended measurements. Feedstock characteristics and pyrolysis conditions will influence biochar homogeneity. These include, but are not limited to; the biomass feedstock type and particle size distribution, pyrolysis temperature and reactor type. The sampling plan specified in Section 8.3.1 of the Biochar Production and Storage Protocol takes a conservative approach to sampling with enough frequency to capture the impacts of any heterogeneity in biochar. These considerations include, but are not limited to, biomass feedstock type and particle size distribution, pyrolysis temperature, reactor type, etc.

The Project Proponent must include all relevant details of their sampling plan, including the number and frequency of sampling and analysis and clear justification of their sampling choice, in the PDD, ensuring compliance with the requirements outlined in Section 8.3.1 of the Biochar Production and Storage Protocol.

4.1.1

Homogeneity Considerations

To ensure representative sampling, composite samples must be divided into a minimum of three representative replicates per batch (although higher replication is recommended), for laboratory analysis, to allow estimation of the mean and standard deviation and detection of potential outliers. Projects must demonstrate the degree of homogeneity within a single Storage or Production Batch. This may include sampling across horizontal and vertical dimensions of a Production or Storage Batch to account for particle sorting that may occur during processing and transportation, as outlined in Section 8.3.1 of the Biochar Production and Storage Protocol. It is the responsibility of Project Proponents to undertake these routine batch characterizations of the biochar utilized within a Crediting Project and detail these in full in the PDD.

Project Proponents must provide a detailed description of how the chosen sampling plan addresses any heterogeneity that might be present within the batch, in the PDD.

4.1.2
Laboratory Requirements

The Project Proponent must report the analytical laboratory/laboratories that have been utilized for biochar analysis and characterization. It is the responsibility of The Project Proponent to ensure that the chosen analytical facilities are reputable and conduct characterization techniques to the required standards indicated within in line with the Protocol and Module. A qualified laboratory is evidenced by accreditation to ISO 17025 or equivalent standards for laboratory quality management for the specific test method. Project Proponents should utilize accredited analytical services such as UKAS, MCERTS, DWTS, and ISO whenever feasible. Where a Project Proponent utilizes laboratory facilities within an academic institution, or a non-accredited commercial laboratory, periodic external validation must be undertaken with an accredited facility. The frequency of these external checks will vary by project and analytical procedure, and will be agreed with Isometric on a case-by-case basis.

Laboratories must complete standard quality assurance procedures on a schedule in accordance with their quality management plans and accreditation requirements to include:

- Instrumentation calibrations and analysis of calibration standards or certified reference materials;
- Analysis of technical replicates, and;
- Analysis of blanks (where possible and appropriate);

4.2
Analytical Checks, Calibration, and QA/QC

If a laboratory is not ISO 17025, or equivalent, accredited, then Project Proponents must:

- Outline specific analytical checks that have been carried out to maintain data quality, with specific reference to the relevant certified reference materials (CRM) used by the utilized laboratory facility.
- Validated of analytical data must be demonstrated through set quality assurance and quality control (QA/QC) criteria within all Crediting Programs.
- All projects must report their QA/QC processes within the PDD, in accordance with the requirements of the Biochar Production and Storage Protocol.

4.3
Data Reporting

Project Proponents are responsible for the delivery of all relevant project data and biochar characterization data to a project's

Validation and Verification Body (VVB), which must be submitted through Isometric's Certify platform. Although a Project Proponent is expected to use external accredited laboratories to produce the data, it is the responsibility of The Project Proponent to deliver data that is accurate and verifiable.

Project Proponents must maintain data records for a minimum of five years following the date of data collection. It is also recommended that a similar approach is taken towards sample archiving, with a representative subsample (e.g. 100 g) dried and archived for a minimum of five years, to allow re-analysis of these, or additional parameters.

Project Proponents must report data such that the data analysis methods used are easily identified, verified and replicated. This Module requires that any data reports include the raw data from which any data analysis and or processing was performed, including reference standards and replicate measurements and any other data associated with quality control and assurance. A summary of the proposed sampling regime must be included in the PDD. Analytical uncertainty, number of samples taken and analyzed, standards used and number of standard runs, standard deviation and percentage error on the standards must also be included in the data report for the VVB.

For example, this may take the form of a spreadsheet containing four dataframes, in all cases appropriate identifiers should be used to allow samples to be easily identified:

1. Summary sheet detailing metadata:
    a. Number of samples run;
    b. Analytical uncertainty;
    c. Standards used;
    d. Number of standards run;
    e. Standard deviation;
    f. Percentage error on standards.
2. Reduced data sheet (data summary);
3. Data processing sheet (if applicable; e.g. processing of ICP-MS data);
4. Raw data;

5.0

Durability of Biochar in Soils

5.1

Quantification of Biochar Durability

5.1.1

Quantification of CO_2e_{stored}

There are two options for calculating the fraction of durable biochar ($F_{durable}$) in this Module. Option 1 results in Credits issued at 200 year durability, and Option 2 results in Credits issued at 1,000 year durability.

In either case, the formula to calculate CO_2e_{stored} is:

$$CO_2e_{stored} = C_{biochar} \times m_{biochar} \times F_{durable} \times \frac{44.01}{12.01}$$

(Equation 1)

Where

- $C_{biochar}$ is the carbon content of the biochar (empirical).
- $m_{biochar}$ is the dry mass of biochar applied.
- $F_{durable}$ is the fraction of durable biochar that remains in the soil for the full duration of the crediting timeline (i.e. either 200 or 1,000 years), and can be credited under this Module.
- $\frac{44.01}{12.01}$ is the mass fraction of carbon dioxide and elemental carbon.

5.1.2

Calculation of C_{org}

This Module only Credits for the durably stored C_{org} fraction in biochar, which is used to calculate CO_2e_{stored}. While biochar associated inorganic carbon generally makes up a small fraction of total biochar C. However, the fate of biochar-associated C_{inorg} is much less predictable in soils[29,30]. Thus, $C_{biochar}$ is calculated using the following equation:

$$C_{biochar} = Total\ Carbon\ Content - C_{inorg}$$

(Equation 2)

Where:

- $Total\ Carbon\ Content$ is the Total Carbon Content of the biochar as analysed using the methods described in **Table 2**.
- C_{inorg} is the inorganic carbon content of the biochar as analysed using the methods described in **Table 2**.

Please refer to Section 8.3.1 of the Biochar Production and Storage Protocol for full guidelines on number of samples required for the measurement of biochar carbon content, $C_{biochar}$.

5.1.2.1 Measurement of Mass of Biochar Applied

Please refer to Section 8.3.1.1 of the Biochar Production and Storage Protocol for full guidelines on measurement of mass of biochar applied, $m_{biochar}$.

5.1.2.2 Calculation of $F_{durable}$

The method of calculating $F_{durable}$ differs between Option 1 (200 year durability) and Option 2 (1,000 year durability). Project Proponents must choose which quantification framework (Option 1, or Option 2, outlined in full below) they wish to use for Crediting their Project. The choice of quantification framework and corresponding durability associated must be clearly outlined in the PDD.

5.1.2.2.1 Option 1: 200 Year Durability

The quantification framework for determining the CO_2e_{stored} for 200-year durability biochar is based on the modelling approach set out by Woolf et al. (2021)[15]. This peer-reviewed paper developed a model to estimate the amount of carbon remaining in soil over time using a decay function. The calculation of $F_{durable}$ requires two inputs: soil temperature at the site of deployment (T_{soil}) and the biochar H/C_{org} ratio. As discussed in Section 1.2, H/C_{org} ratio is a proxy measurement of the (poly)aromaticity of a biochar sample[31]. Additionally, soil temperature is highly related to microbial and chemical degradation of biochar with higher temperatures potentially increasing mineralisation rates[9].

As such, the formula to calculate Fdurable for 200 year durability is:

$$F_{durable,200} = \min\Big(0.95,\ 1 - \Big[c + (a + b \cdot \ln(T_{soil})) \cdot H/C_{org}\Big]\Big)$$

(Equation 3)

Where

- $F_{durable,\,200}$ is the fraction of carbon remaining in durable storage after 200 years.
- a, b, and c are estimated parameters based on an analysis of the data of Woolf (2021)[15], described in more detail below.
- T_{soil} is the average annual temperature of the soil where the biochar is stored (°C).
- H/C_{org} is the molar hydrogen-organic carbon ratio.

The parameters a, b, and c are fixed for the time horizon of 200 years estimated using data available in the appendix of Woolf et al. (2021)[15]. Parameters theoretically could be recalculated for longer or shorter time horizons for Crediting according to the method. Additionally, the model is calibrated using data from biochars with H/C_{org} ratios above 0.1. For biochars with higher degrees of carbonization (H/C_{org} below 0.1), the maximum durable fraction is conservatively capped at 95% to reflect expected mineralization of the labile fraction and allow for some minor degradation of the stable fraction. Temperature data should be submitted to a maximum of one decimal place, and a conservative minimum threshold of 7°C is applied to account for uncertainties regarding soil temperature influences in locations subject to periodic soil freezing.

To ensure a conservative estimate of the decay function, the coefficients a, b and c are calculated using the 17th percentile of durability distribution, roughly equivalent to one standard deviation below the mean, assuming normal distribution. The parameters are estimated using a two-stage regression analysis summarized below.

- Stage 1 - Quantile regression
  - A linear quantile regression of the non-durable portion (1 – $F_{durable}$) on H/C_{org} ratio was conducted for each unique soil temperature in the Woolf et al. (2021)[15] dataset.
    - Quantile regression was used to estimate the value of 1 – $F_{durable}$ corresponding to the 17th percentile of the durability distribution, conditional on the H/C_{org} and T_{soil}. This percentile was chosen as a conservative estimate of biochar carbon durability.
- Stage 2 - Deriving parameter
  - The intercepts from the regressions (one for each soil temperature) were regressed on a constant to obtain parameter c.
  - The slopes (coefficients on H/C_{org}) from the first-stage regressions were regressed against the natural logarithm of soil temperature $\ln(T_{soil})$ in a linear regression. This provided parameters a and b, which capture how soil temperature modifies the relationship between H/C_{org} and (1 – $F_{durable}$). The resulting parameters a, b, and c are (summarised in **table 4**) are used to calculate the 200 year carbon durability.

Table 4: The fixed, conservative coefficients of biochar decay calculated from Woolf et al., (2021)[15] calculated for Equation 3, for the time horizon of 200 years for crediting, using data available in the appendix of Woolf et al., (2021)[15].

PARAMETER	VALUE
a	-0.383
b	0.350
c	-0.048

Project Proponents must provide T_{soil} data either by:

- Baselining their own annual soil temperature measurements, in order to ensure that the data used in the calculation for crediting comes from direct, project-specific measurements. These must be carried out according to ISO 4974, or equivalent, and justified in the PDD. A dataset of measurements for T_{soil} taken for the year preceding crediting, and the average T_{soil} calculated from that dataset, must be included in the PDD. Project Proponents must report the average of monthly soil temperature measurements from every application site. At least 10 measurements must be taken per site-month.

- Or, if no baselining data for soil temperature is available, a justifiable T_{soil} for calculation of the durable fraction of biochar must be obtained from a global database of soil temperatures such, as Lembrechts et al. (2022)[32], or equivalent. Project Proponents must identify which region their Project best aligns with from the global dataset, and justify both the dataset used and the average annual soil temperature chosen in the PDD. Air temperatures must not be used as a proxy for average annual soil temperatures. While average air and soil temperatures are correlated, there is evidence that mean annual soil temperatures can be 2-4°C warmer than mean annual air temperatures.

Additional T_{soil} requirements:

- Projects must justify that the same carbon removal quantification (i.e., T_{soil}) can reasonably be applied across the entire project area.
- If the soil temperature variation within a project boundary exceeds 1°C then The Project must be further divided, or the most conservative temperature value (i.e., highest) within the project area must be used for the quantification of $F_{durable}$. Details on the size of the project boundary chosen and variability of soil temperature within the project boundaries should be outlined in the PDD.

- The VVB and/or Isometric may request additional documentation explaining how individual sites were selected to fit within the project boundary.

5.1.2.2.2 Option 2: 1,000 Year Durability

The quantification framework for determining the CO_2e_{stored} for 1000-year durability is based on the quantification approach set out by Sanei et al., (2024)[11] This approach quantifies biochar on the random reflectance value of the biochar, compared to inertinite as a proxy for geologically stable carbon. Using petrographic analysis, Sanei et al., (2024)[11] identified that biochars with a mean random reflectance ($R_0 \geq 2\%$) exhibit structural characteristics equivalent to inertinite macerals in fossil coals and chars, which are known to persist over geological timescales. While biochar meeting the benchmark of $R_0 \geq 2\%$ can be considered permanent, additional peer-reviewed research[33,34,35] has been published that further validates the experimental work of Sanei et al., (2024)[11].

As outlined in Section 3 of this Module, Project Proponents must report a set of at least 500 measurements of R_0, calculated at the maceral-level, for at least three replicate samples of their biochar. Batches that adopt this measurement approach can be credited for the percentage of their biochar which passes the 2% R_0 benchmark, as outlined in Sanei et al. (2024)[11]. The histogram of the $_0$ values must be submitted at the point of project verification for this Crediting option. This method was further updated in Sanei et al., (2025)[36] to refine the methodology to only account for the recalcitrant fraction of biochar (discounting the reactive fraction, determined by thermogravimetric analysis).

To ensure a conservative approach when Crediting biochar durability, we account for uncertainty in both R_0 measurement, and the proportion of carbon that is non-reactive. Specifically, the credited durable fraction ($F_{durable}$) is calculated using the mean values for each parameter reduced by one standard deviation. This method ensures that the durability estimate reflects a lower-bound confidence level, mitigating the risk of overestimating long-term carbon storage.

As such, $F_{durable}$ is calculated using through:

Calculating the sample standard deviation of R_0 quantifying the typical deviation of individual R_0 measurements from their mean, which is used to account for uncertainty and conservatively adjust the estimated durable fraction of biochar carbon, as:

$$s_{R_0} = \sqrt{\frac{1}{n_{R_0} - 1} \sum_{i=1}^{n_{R_0}} (R_{0,i} - \bar{R}_0)^2}$$

(Equation 4)

Where:

- s_{R_0} is the standard deviation of R_0 measurements
- n_{R_0} is the number of measurements in the sample (i.e. >3).
- $R_{0,i}$ is the individual measurement of for the i-th sample.
- $\bar{R}_0$ is the mean of all R_0 measurements.

Calculating the sample standard deviation of $C_{non-reactive}$ quantifying the typical deviation of individual $C_{non-reactive}$ measurements from their mean, which is used to account for uncertainty and conservatively adjust the estimated durable fraction of biochar carbon, as:

$$s_{C_{non-reactive}} = \sqrt{\frac{1}{n_{C_{non-reactive}} - 1} \sum_{i=1}^{n_{C_{non-reactive}}} \left(C_{\text{non-reactive},i} - \bar{C}_{\text{non-reactive}}\right)^2}$$

(Equation 5)

- $s_{C_{non-reactive}}$ is the standard deviation of $C_{non-reactive}$ measurements
- $n_{C_{non-reactive}}$ is the number of measurements in the sample (i.e. >3).
- $n_{C_{non-reactive}}$ is the individual measurement of for the i-th sample.
- $\bar{C}_{\text{non-reactive}}$ is the mean of all $C_{non-reactive}$ measurements

Then:

$$F_{\text{durable},1000} = \min\left(0.95, \max\left(0, (\bar{R}_0 - s_{R_0})(\bar{C}_{\text{non-reactive}} - s_{C_{non-reactive}})\right)\right)$$

(Equation 6)

Where:

- $F_{durable,\,1000}$ is the fraction of durable (inert) carbon in the biochar after 1000 years, adjusted conservatively for uncertainty.

- $\bar{R}_0$ is the mean of all R_0 measurements.
- s_{R_0} is the standard deviation of R_0.
- $\bar{C}_{\text{non-reactive}}$ is the mean of all non-reactive carbon measurements.
- $s_{C_{non-reactive}}$ is the standard deviation of $C_{non-reactive}$.

The maximum and minimum functions are applied to ensure that the fractions are bounded.

5.1.3
Combining Durability Options

Project Proponents may choose to issue carbon Credits using a combination of both the 200-year and 1000-year durability Crediting option from a single facility under a unified project validation.

Project Proponents must state if they are opting to pursue this combined durability option in the PDD.

Each durability tier must undergo a separate verification assessment, reflecting the distinct evidence and durability requirements for each Crediting option (see Sections 4.1.1.3.1 and 4.1.1.3.2 for analytical requirements for the 200- and 1000-year durability options respectively).

To maintain clear traceability and prevent double counting, each production batch must be credited under only one durability tier. If storage batch mixing is proposed for production batches from a single production process, the Proponent must provide a clear plan for maintaining batch-level traceability and ensuring that durability-specific risks, such as combustion or surface soil disturbance, do not compromise credit integrity. In order to ensure standards are maintained, Project Proponents are required to separate production batches by both of the following:

- **Temporal separation**: The Project Proponent must ensure production batches are distinctly separated by time. This may be evidenced by documentation of specific production batches for each durability tier (e.g., dates during which all facility output is credited as 200-years vs. 1000-years).

- **Spatial separation**: The Project Proponent must document distinct separation of stockpiling (if applicable) and storage locations, with traceable batch labeling and chain-of-custody documentation.

These protections must be documented in the PDD and agreed upon with Isometric prior to credit issuance. If Isometric or the appointed VVB identify any issues or discrepancies with the traceability of combined durability options, then credits will be capped at 200 year durability.

5.2
Environmental Monitoring

To ensure the long-term durability of carbon stored through biochar application, Project Proponents should track relevant environmental and management conditions that may influence carbon stability. While biochar itself is chemically stable, surrounding environmental factors, particularly land management practices, can affect both direct carbon

losses (e.g., through enhanced decomposition) and indirect emissions (e.g., through soil disturbance or changes in nutrient cycling). This Section outlines the key parameters that should be monitored and reported to assess environmental conditions over time, mitigate reversal risks, and ensure the integrity of credited CO_2 removals across a range of land use contexts.

5.2.1 Site Management

Land and field management practices can influence the durability of carbon removal in both direct and indirect ways[37,38,39]. For example, irrigation could significantly impact both moisture and pH, and soil moisture has been shown to have an impact on biochar degradation rate[14]. Furthermore, soil tillage can lead to increased carbon flux in the topsoil, which can affect SOC stocks[40]. While some of these practices are most commonly associated with agriculture, similar interventions are found in forestry, land restoration, and other managed soil systems. Project Proponents should assess and report all relevant practices as part of the Greenhouse Gas (GHG) Statement, regardless of land use type.

The following site management parameters should be submitted within the GHG statement, where applicable:

- Irrigation schedule
- Irrigation source
- Tillage or soil disturbance practice
- Fertilizer or amendment usage
- Fertilizer/amendment composition
- Vegetation type and management
- Pre-deployment, deployment, and post-deployment monitoring

This Section outlines the approach Project Proponents should take to track site management practices during the Crediting Period. Proponents should ensure that the application of biochar does not lead to material changes in field or site management that could result in additional CO_2e emissions. Any shifts in practices, such as changes to tillage, irrigation, or fertilizer use, should be carefully assessed and reported to confirm that the project does not introduce unintended emissions that compromise the net carbon benefit.

5.2.2 Baseline Establishment

Project Proponents should establish a baseline of soil conditions prior to biochar application in order to (i) verify CO_2 sequestration attributable to project activities, and (ii) enable ongoing assessment of potential environmental impacts. Baseline soil samples should be collected before biochar is applied, and must characterize

heterogeneity in key soil parameters relevant to biochar carbon removal, including pH, soil texture, moisture content, and SOC. If sampling is undertaken, sampling must be conducted during a defined seasonal window, and this timing must be kept consistent for any future sampling rounds to ensure comparability over time and account for seasonal variability in SOC stocks. The sampling strategy must address spatial heterogeneity across the project site, and the full methodology must be clearly described and justified in the PDD.

To minimize sampling bias, soil should be collected to the maximum tillage depth or 30 cm, whichever is deeper. While random sampling Protocols are preferred, alternative sampling designs may be used if they are fully documented and justified in the PDD. All baseline samples must be analyzed for the parameters listed in Table 4.

Table 5: Parameters for measurement to be used during Project baseline establishment.

PROPERTY	EXPECTED UNIT	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Soil pH	unit less	-	pH measurement in soil slurry e.g. ISO 10390:2021	Level of acidity or alkalinity of soil. pH controls nutrient availability, microbial activity, and heavy metal mobility; most nutrients are optimally available in neutral to slightly acidic soils (pH 6.0-7.0). Extreme pH levels can make essential nutrients unavailable to plants or cause toxic elements to become soluble, directly impacting plant growth and soil biological processes.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended
Soil moisture content	wt.%	-	Determination of water content in soils e.g. ISO 17892-1:2014	Moisture content of soil to which the biochar will be applied. Soil moisture regulates plant water availability, nutrient transport, and microbial activity; optimal moisture levels support root function, facilitate nutrient uptake through soil solution, and maintain the biological processes essential for organic matter decomposition and nutrient cycling.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended
Bulk density	g cm^{-3} or kg m^{-3}	-	Determination of dry bulk density e.g. ISO 11272:2017	Bulk density indicates the level of soil compaction and pore space availability, affecting root penetration, water infiltration, and air movement. Higher bulk density values suggest compacted soils that restrict root growth, reduce water and nutrient uptake, and limit oxygen availability for plant roots and soil microorganisms, ultimately impacting overall soil productivity and health.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended

PROPERTY	EXPECTED UNIT	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Soil type and texture		-	Oven drying coupled with gravimetric sieving, Laser diffraction or x-ray scattering e.g. ISO 11277:2020	Type of soil quality (sand, silt, clay) and the size distribution. These determine fundamental soil properties including water retention, drainage, nutrient holding capacity, and workability. Clay soils retain more nutrients and water but may have drainage issues, while sandy soils drain well but have lower nutrient retention. Understanding texture helps predict soil behavior, management needs, and potential limitations for plant growth and agricultural practices.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended
Nutrient availability		-	Characterizing nutrient availability should involve testing electrical conductivity (EC) and calculating the total dissolved solids (TDS) content of soil leachates with a commercial water quality test meter.	Nutrient availability is a measure of the concentration of essential plant nutrients (nitrogen, phosphorus, potassium, and micronutrients) in forms that plants can readily absorb. This directly impacts plant growth, crop yields, and ecosystem productivity. Nutrient deficiencies limit plant development, while excess nutrients can cause toxicity, environmental pollution through runoff, and disruption of soil microbial communities.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended
Soil Organic Carbon (SOC)	g kg^{-1} or ppm	-	Dry combustion, Walkley-Black method e.g. ISO 10694:1995	SOC serves as the primary indicator of soil organic matter and overall soil health. It enhances soil structure, water retention, and nutrient cycling while supporting diverse microbial communities. Higher organic carbon levels indicate better soil fertility, improved resilience to environmental stress, and greater capacity for long-term agricultural productivity.	Only at project validation. A sufficient number of samples should be taken that are representative of conditions in the application site. Details of the sampling regime chosen should be outlined in the PDD.	Recommended
Crop yield or farm profitability data for three years proceeding biochar application	Various accepted	-	Various accepted	This data allows baseline establishment for The Project. Serves as the ultimate measure of soil health effectiveness, integrating all physical, chemical, and biological soil factors into economic outcomes. This parameter validates that soil health improvements translate into tangible benefits for farmers and food security while indicating the long-term viability of current management practices.	Only at project validation. Details of the type of data submitted must be outlined and justified in the PDD.	Recommended

5.2.2.1

Site Selection to Minimizing Reversal Risks

Beyond the biomass feedstock type and the physical and chemical characteristics of biochar, its durability is also influenced by environmental and anthropogenic factors. Environmental conditions primarily affect the degradation of the labile (less stable) fraction of biochar, while the recalcitrant (more durable) fraction will remain stable in soil throughout the designated durability window.

Project Proponents should carefully evaluate potential sites based on key environmental factors to minimize negative impacts on biochar durability, migration, and overall soil and plant health.

Environmental factors:

- Climate:
  - Biochar should not be spread on land that:
    - Has been frozen for 12 hours or more in the preceding 24 hours.
    - Is waterlogged, frozen or covered in snow.
  - Extreme fluctuations in soil temperature, such as freeze-thaw events.
- Soil conditions:
  - Soil texture: fine and coarse grained biochars are more likely to have larger impacts on soil characteristics than medium grained[41].
  - Mean soil temperature: higher soil temperatures increase degradation speed of biochar degradation, particularly the reactive fraction[9].
  - Nutrient availability: higher nutrient availability in soil could potentially impact microbial activity.
  - Water management (e.g. irrigation source and schedule, drainage modification or hydrological restoration).
  - Topography and geography:
    - Biochar must not be spread on steep slopes where there is a significant risk of loss through erosion, unless evidence can be provided to show sufficient preventative measures to protect against erosion.
    - Biochar should not be stored or spread within 10 meters from any watercourse and 50 meters from any spring, well or borehole.

Anthropogenic factors:

All of the following activities taking place on agricultural land may impact the environmental factors above, and therefore impact biochar degradation.

- Irrigation source and schedule: as above, high soil moisture decreases biochar MRTs.
- Fertilizer use and composition: this could alter nutrient availability in soils, and by extension, microbial activity.
- Crop type and rotation: similar to the above discussion on root growth, different crops may interfere with mechanical breakdown processes, soil pH and/or soil moisture content.
- Land management practices such as tilling, plowing, seeding, and harvesting.

Project Proponents should outline in the PDD a full description of site conditions, including justification of how site suitability was chosen bearing in mind the factors listed above.

6.0

Proof of Biochar End Use

Full details of biochar post-production processing and movement must be included in the PDD (in addition to all others listed in this Module), in order to evidence that biochar storage has occurred.

Extensive guidance on the acceptable evidence to ensure that storage of biochar will occur in the soil environment and risk of reversal is minimized is included in Appendix II. This Appendix recognizes the diversity of routes through which biochar may be applied to soil and provides a comprehensive list of the types of evidence that are appropriate for each pathway.

In all cases, Biochar must be applied at an appropriate moisture level to minimize dust loss, which can be damaging to human health and the wider environment, and prevent negative effects on soil biology.

In all instances evidence should comply with the Isometric Standard principals for transparency, and biochar should be traceable to the end user or retailer.

6.1

Calculation of $CO_2e_{Emissions,RP}$

$CO_2e_{Emissions,RP}$ is the total greenhouse gas emissions associated with a given Reporting Period (RP).

Equations and emissions calculation requirements for including emissions associated with reactor operations and reaction monitoring, are set out in the Protocol and are not included in this Module. Specific considerations for CO_2 stored as biochar in soils are set out here.

This section specifically refers to biochar that is mixed with compost. Transport emissions must be considered if the average transport distance for biochar-amended compost is more than the average transport distance of normal products produced by the production facility. All other downstream emissions may be excluded from the system boundary if these activities were already occurring and would continue to occur in the absence of The Project. This can be evidenced by providing documentation that biochar-amended product meets the same performance requirements of a conventional product for the intended use case. Further information can be found in Appendix II.

7.0

Buffer Pool and Reversal Risk

7.1

Buffer Pool

As outlined in Section 2.5.9 of the Isometric Standard, the Buffer Pool provides insurance against Reversal risks that may be observable and attributable to a particular project through monitoring. In the case of biochar, such reversals are not expected to be directly measurable or attributable to a particular project. Projects Crediting against this Protocol are credited conservatively to account for degradation of labile pools of biochar within the relevant crediting time horizon. Projects adhering to the Module are categorized as having a Very Low Risk of Reversal, and therefore contribute 2% of issued Credits to the Buffer Pool as a precaution against residual uncertainties.

Following the Section 2.5.9 of the Isometric Standard, storage uncertainty for open systems is primarily accounted for within the removal quantification framework. For more details on Reversals, refer to Sections 2.5.9 and 5.6 of the Isometric Standard.

7.2

Risk of Post-Deployment Use as Fuel

Project Proponents must assess and disclose the risk that biochar, once applied to soils, may be deliberately removed and used as a fuel source.

Project Proponents must assess and disclose the risk that biochar, once applied to soils, may be deliberately removed and used as a fuel source.

This assessment should consider the local context, including energy demand, access to fuel sources, economic incentives for removal, and the visibility or accessibility of biochar post-deployment.

Projects facing a non-negligible risk of post-deployment removal via picking of biochar from the application site must ensure that the majority of biochar applied has a particle size of ≤ 10 mm, with at least 95% by weight passing through a 10 mm sieve. Larger particle sizes (e.g., > 20 mm) should be avoided unless fully incorporated into the soil immediately on application or otherwise rendered inaccessible.

Project Proponents where there is a material risk of collection and use as a fuel must report the particle size distribution and provide justification that the biochar will not be collected or used for combustion.

7.3

Biochar Stockpiling Duration and Safety

Biochar may be stockpiled between production and reaching its end use.

Project Proponents must confirm whether they stockpile biochar, post production, pre-end use.

However, biochar stockpiling requires careful management to maintain material integrity and prevent losses.

Biochar must not be stored for more than 12 months after production, unless otherwise agreed upon with Isometric.

During this period, every precaution must be taken to ensure no biochar is lost through environmental factors or handling.

As such, the biochar must:

- Be stored in a wet condition or mixed with moist compost to minimize the risk of self-ignition.
- Be stored in a covered environment, ideally indoors, but at minimum under a secure, weatherproof cover that protects the material from rain, wind, and other environmental conditions.
- Be stored at a location that avoids proximity to waterways such as rivers, streams, or drainage areas where runoff could result in biochar loss.

Proper storage Protocols are essential to preserve the biochar's carbon sequestration potential and ensure its effectiveness for subsequent soil application.

All details of the stockpiling location and mitigation methods put in place to ensure the biochar is stable during stockpiling, must be detailed in the PDD.

If biochar is reversed during stockpiling the carbon associated with the creation of that biochar must still be counted within the project boundary, despite it not making it to storage.

8.0

Appendix I: Overview of Required Measurement to Credit Using This Module

Please note, all data should be reported in the International System of Units (i.e., SI, metric) to avoid confusion in calculation.

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
NET CDR QUANTIFICATION							
CHEMICAL AND PHYSICAL CHARACTERIZATION							
$Total\ Carbon\ Content$	The carbon content of applied biochar is necessary for the calculation of C_{org} and thus CO_2e_{stored}, in accordance with Section 8.3 of the Biochar Production and Storage Protocol. See Section 8.3.1 of the Biochar Production and Storage Protocol for carbon content sampling guidance.	Required	-	Measured	% (weight / weight)	Eq. 2 in the Biochar Production and Storage Protocol	ISO 29541:2025 or A D5373-21
$m_{biochar}$	Total mass of applied biochar used to calculate the dry mass of CO_2e_{stored}	Required	-	Measured	metric tonnes	Eq. 2 and 3 in the Biochar Production and Storage Protocol	Direct mass measurement
Moisture Content	The moisture content of applied biochar is necessary for the quantification of CO_2e_{stored}, in accordance with Section 8.3.1 of the Biochar Production and Storage Protocol. Carbon content is be reported on a dry basis to account for differences in total biochar mass.	Required	-	Measured	% (weight / weight)	Eq. 2 and 3 in the Biochar Production and Storage Protocol	Analytical measurer of moisture content biochar
Inorganic Carbon Content (C_{inorg})	Measurement of C_{inorg} in biochar is required to	Required	-	Measured	% (weight / weight)	< Section 3.3 in the Biochar Storage in Soil	Analytical measurer of inorganic carbon

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	accurately differentiate organic carbon C_{org} from $Total\ Carbon\ Content$, which may include both inorganic and organic forms. Only C_{org} is credited for under this Protocol and Module.					Environments Module	content of applied biochar
Total Hydrogen (H)	Measurement of H is required to calculate the H/C_{org} ratio.	Required	-	Measured	% (weight / weight)	Eq. 3 in the Biochar Storage in Soil Environments Module	Analytical determina of hydrogen conten biochar, used to calculate biochar
Total Nitrogen (N)	Nitrogen is a key component that influences biochar's properties and its potential applications, including its use as a soil amendment.	Required	-	Measured	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of nitrogen content biochar
Total Oxygen (O)	Measurement of O is required to calculate the O/C_{org} ratio, which is an addition metric of stability, used to confirm the H/C_{org} ratio.	Required	-	Measured	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of oxygen content in biochar
Total Sulfur (S)	Sulfur is a key component that influences biochar's properties and its potential applications, including its use as a soil amendment.	Required	-	Measured	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of sulfur content in biochar
Organic carbon (C_{org}) Content	C_{org} is derived from the total carbon content minus the inorganic carbon content in the sample. C_{org} represents the initial total of organic carbon stored in biochar.	Required	-	Calculated	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module | $Total\ Carbon\ Content$ and C_{inorg}.	$Total\ Carbon\ Co$ and C_{inorg}.

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	This is the basis on which CO_2e_{stored} is calculated taking into account the mass of biochar applied and the durability of the carbon.						
Molar H/C_{org} ratio	Molar H/C_{org} is derived from the H and C_{org}. Low H/C_{org} ratios indicate the presence of significant amounts of aromatic compounds within the biochar, which are highly stable and conducive to long-term stability of sequestered biochar in soil. For the 200 year crediting option, this is used to model biochar durability.	Required	<0.5	Calculated	Ratio	Eq. 3 in the Biochar Storage in Soil Environments Module	H and C_{org}
Molar O/C_{org} ratio	Molar O/C_{org} is derived from the O and C_{org}. The O/C_{org} ratio indicates the presence of functional groups, with lower ratios indicative of fewer functional groups. A lower abundance of functional groups is favorable for biochar permanence, as these groups can serve as reactive sites on the biochar surface and potentially enhance degradation processes. C-O bonds are more labile than C-C bonds. Furthermore, the O/C_{org} ratio is required to verify that low O/C_{org} ratios genuinely reflect a high degree of aromaticity, rather than the presence of oxygenated aliphatic carbon.	Required	<0.2	Calculated	Ratio	Section 3.3 in the Biochar Storage in Soil Environments Module	O and C_{org}

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
Ash content	Measurement of ash content in biochar is important because it represents the inorganic, non-combustible fraction remaining after complete combustion. Ash content can influence soil pH, nutrient availability, and biochar's capacity to retain water and nutrients when applied to soil.	Required	-	Measured	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of ash content in bio
Bulk density (<3 mm particle size)	This measurement standardizes particle size to <3 mm to provide a consistent metric for comparing different biochar samples. It is primarily used for research and characterization purposes, as it eliminates the variability caused by particle size distribution. Bulk density of the <3 mm fraction also provides insights into the porosity and compaction characteristics of the finer material.	Required	-	Measured	kg m^{-3}.	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of bulk density of th mm fraction of bioc
Volatile matter content (VMC)/ Volatile Compounds	VMC is indicative of the level of carbonisation, stability, and reactivity of biochar. A higher VMC suggests greater reactivity, while a lower VMC means reduced interaction with soil components.	Recommended	-	Measured	% (weight / weight)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of level of carbonisa stability, and reactiv biochar
pH	Biochar pH reflects its potential impact on soil health, quality, and microbial activity when used as a soil amendment. Measuring	Required	-	Measured	-	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of pH

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	pH will assess biochar's influence on these factors. Additionally, pH indirectly affects biochar durability. However, there is no specific eligibility threshold for biochar pH.						
Salt content	Salt content is an important parameter in biochar characterization because elevated levels of soluble salts can negatively affect soil health and plant growth when the biochar is applied.	Required	-	Measured	g kg^{-1}	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of salt content
Water Holding Capacity (WHC)	Water holding capacity (WHC) is an important property of biochar because it influences soil moisture retention, plant-available water, and overall soil structure when the biochar is applied.	Required	-	Measured	%	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical measurer of WHC
Declaration of the nutrient content (P, K, Mg, Ca, Fe)	Declaration of the nutrient content of biochar is important because these elements (phosphorus, potassium, magnesium, calcium, and iron) contribute to soil fertility and plant nutrition when biochar is applied.	Required	-	Measured	g kg^{-1}	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of nutrient content
Heavy metals (lead (Pb), cadmium (Cd), copper (Cu), nickel (Ni), mercury (Hg), zinc (Zn), chromium (Cr), and arsenic (As)	Quantification of heavy metals in biochar is essential to ensure environmental and human health safety. Elevated concentrations of metals such as lead (Pb), cadmium (Cd), copper (Cu), nickel (Ni), mercury (Hg), zinc (Zn), chromium (Cr), and arsenic (As) can pose	Required	Pb = 300 g $^{-1}$ DM, Cd = 5 g t^{-1} DM, Cu = 200 g t^{-1} DM, Ni = 100 g t^{-1} DM, Hg = 2 g t^{-1} DM, Zn = 1000 g t^{-1} DM, Cr = 200 g t^{-1}	Measured	mg kg^{-1} or g t^{-1} DM (directly equivalent)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical measurer of heavy metal cont

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	risks of soil and water contamination, bioaccumulation in crops, and potential toxicity to soil organisms. Measuring and declaring heavy metal content allows for verification against regulatory limits and safeguards the suitability of biochar for soil application.		DM, As = 20 g t^{-1} DM				
Polycyclic aromatic hydrocarbons (PAHs) - U.S. Environmental Protection Agency (EPA) 16 and European Food Safety Authority (EFSA) 8	Measurement of PAHs in biochar is required to assess potential environmental and human health risks. PAHs are a group of organic contaminants that can form during pyrolysis, and some are known to be carcinogenic or otherwise toxic. The EPA 16 set refers to the 16 priority PAHs identified by the U.S. EPA, while the EFSA 8 subset refers to the eight PAHs prioritized by the EFSA for food and feed safety. Quantifying these compounds ensures that biochar complies with international safety standards and is suitable for soil application.	Required	EPA 16 = declaration, EFSA 8 = 1 g $^{-1}$ DM	Calculated	mg kg^{-1} or g t^{-1} DM (directly equivalent)	Section 3.3 in the Biochar Storage in Soil Environments Module	Calculated from DIN 17503
Polychlorinated dibenzodioxins/-furans (17 PCDD/F)	easurement of the 17 toxicologically relevant polychlorinated dibenzodioxins and dibenzofurans (PCDD/F) is required because these persistent organic pollutants can form as by-products during the pyrolysis of certain feedstocks. PCDD/F compounds are highly	Required	PCDD/F: 20 ng kg-1 DM	Measured	ng kg^{-1} DM	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determin of PCCD/F compou

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	toxic, bioaccumulative, and can pose significant risks to human health and the environment. Quantifying their levels ensures that biochar complies with international safety limits and is suitable for soil application without introducing harmful contaminants.						
Polychlorinated biphenyl (12 World Health Organization (WHO) PCBs)	Measurement of the 12 dioxin-like polychlorinated biphenyls (WHO-PCBs) is required because these compounds are toxic, persistent, and can bioaccumulate in the environment. They may be introduced through contaminated feedstocks or form as trace by-products under certain production conditions. Quantifying WHO-PCBs ensures that biochar meets international safety standards, safeguarding soil health, food chains, and human health when applied to land.	Required	PCB: 0.2 mg kg^{-1} DM	Measured	mg kg^{-1} DM, sometimes reported in µg kg^{-1} DM (to convert divide by 1000)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina of PCB compounds
Bulk Carbon Bonding State	High aromaticity and aromatic condensation are shown to increase MRT by an order of magnitude. High degrees of aromatic condensation result in biochar that is less prone to microbial activity.	Recommended	-	Measured	% by carbon bonding type (aromatic, aliphatic, carbonyl)	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina using NMR spectroscopy
External surface carbon bonding state composition	Biochar degrades from the outside in. If the exterior of the biochar particles has a different chemical than the center,	Recommended	-	Measured	Relative proportion (%) of each functional	Section 3.3 in the Biochar Storage in Soil Environments Module	Analytical determina using XPS analysis o pulverised and unpulverised sampl Comparing external

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	that affects degradation rate. Comparing external to internal composition without depth profiling can be done by comparing XPS of in-tact particles to Raman/NMR of pulverised samples OR XPS of pulverised and unpulverised samples. In either case, the sample preparation should be specified in the PDD. Pulverising samples ensures the average chemical composition throughout the particle is measured, whereas the surface composition of in-tact particles can be characterised by XPS.				group out of the total surface carbon detected		internal compositio without depth profil can be done by comparing XPS of i particles to Raman/ of pulverised sampl OR XPS of pulverise and unpulverised samples. In either c the sample prepara should be specified the PDD.
Gross calorific value	Indicator of energy content of biochar	Recommended	-	Measured	kJ kg^{-1}		Analytical determin of the total amount heat released when sample is complete combusted in an ox rich environment
Net calorific value	Indicator of the energy content of biochar	Recommended	-	Measured	kJ kg^{-1}		Analytical determin of the amount of he released by the co combustion of a sa in an oxygen-rich environment exclu the latent heat of vaporization of wat formed during combustion.
Specific surface area	Surface area of biochar applied to soils	Recommended	-	Measured	m^2 g^{-1}	Section 3.2 in the Biochar Storage in Soil Environments Module	Measurement of the surface area of bioc applied to soils

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
Porosity	Percentage of void space in biochar	Recommended	-	Measured	%	Section 3.2 in the Biochar Storage in Soil Environments Module	Analytical determin of the total void spa in biochar, an indica water adsorption potential
Particle size distribution	Estimate of the range and proportion of different sized particles within a biochar sample	Recommended	-	Measured	% by size fraction	Section 3.2 in the Biochar Storage in Soil Environments Module	Measurement of the range and proportic different sized parti within a biochar san
ADDITIONAL REQUIREMENTS FOR 1,000 YEAR DURABILITY ONLY							
Random reflectance (R_0)	Random reflectance is an indicator of aromaticity, aromatic ring unit size and condensation. A R_0 value greater than 2% has been proposed as a benchmark for quantifying the permanent pool of carbon in a biochar [11]. The R_0 frequency distribution histogram is used to decide what fraction of biochar above this benchmark can be classified as geologically inert [11]. This is used to calculate $F_{inert,1000}$.	Required	> 2 % for inertinite (creditable fraction)	Measured	%	Eq. 4 in the Biochar Storage in Soil Environments Module	Analytical determin of random reflectan R0 measurements (500 measurements) different biochar macerals
Reactive Organic Carbon and Residual Organic Carbon ($C_{non-reactive}$)	Measurement of reactive organic carbon in biochar is important because this fraction represents the more labile, easily degradable component of organic carbon. Elevated levels of reactive organic carbon can reduce biochar's long-term carbon stability, as it is more susceptible to	Required	-	Measured	%	Eq. 4 in the Biochar Storage in Soil Environments Module	Re-pyrolysis of bioc to achieve separatic the reactive and res organic carbon pha

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
	microbial decomposition and mineralization in soil. Random reflectance values are subsequently only applied to the residual, stable fraction of biochar. This is used to calculate $F_{inert,1000}$.						
REACTOR MONITORING							
m_i	Mass flow rate of gas emitted to atmosphere from biomass pyrolysis	Required	-	Measured or calculated	tonnes hr^{-1}	Eq. 7 in the Biochar Production and Storage Protocol	Direct flow measurements Emis testing results Mate balance calculation
$C_{Tailgas,CO2}$	Concentration of CO_2 in tailgas as a mass fraction	Required	-	Measured	wt.%	Eq. 7 in the Biochar Production and Storage Protocol	On - line analyzer Emissions testing re

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
$C_{Tailgas,CH4}$	Concentration of CH_4 in tailgas as a mass fraction	Required	-	Measured	wt.%	Eq. 7 in the Biochar Production and Storage Protocol	On - line analyzer Emissions testing re
$C_{Tailgas,CO}$	Concentration of CO in tailgas as a mass fraction	Required	-	Measured	wt.%	Eq. 7 in the Biochar Production and Storage Protocol	On - line analyzer Emissions testing re

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
$m_{fuel,conversion}$	Mass of fuel used in biomass conversion	Required	-	Measured	liters	Eq.5 of Energy Use Accounting Module	Fuel usage records
$EF_{Fuel,conversion}$	Fuel emission factor for biomass conversion	Required	-	Estimated	CO_2e unit $(tonnes)^{-1}$	Eq.5 of Energy Use Accounting Module	Argonne National Laboratory GREET Model, California Ai Resources Board modified GREET mo (CA-GREET), Ecoinv database, US Feder Life Cycle Inventory database or LCA Commons, or from s databases used in common LCA practi or tools
$kwh_{Conversion}$	Electricity usage for biomass conversion	Required	-	Measured	kWh	Eq. 2 Energy Use Accounting Module	Electricity usage rec
$EF_{Elect,Conversion}$	Electricity emission factor	Required	-	Estimated	CO_2e kwh $(tonnes)^{-1}$	Eq. 2 Energy Use Accounting Module	Argonne National Laboratory GREET Model, California Ai Resources Board modified GREET mo (CA-GREET), Ecoinv database, US Feder Life Cycle Inventory database or LCA Commons, or from s databases used in common LCA practi or tools
$EF_{Fuel,Transportation}$	Fuel emission factor for transportation	Required Under certain conditions	-	Measured or Estimated	CO^2e unit $(tonnes)^{-1}$	Eq. 2 of the Transportation Emissions Accounting Module	Argonne National Laboratory GREET Model, California Ai Resources Board

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
							modified GREET mo (CA-GREET), Ecoinv database, US Feder Life Cycle Inventory database or LCA Commons, or from s databases used in common LCA practi or tools
$W_{Conversion}$	Mass of biomass transported from supplier to conversion site	Required Under certain conditions	-	Measured	kg, tonne	Eq. 2 of the Transportation Emissions Accounting Module	Shipping records (b lading) OR Fleet management recorc Weighscale tickets
Product Stage Emissions	Includes raw material sourcing, transport to facility and manufacturing	Required	-	Measured	tonnes	Section 8.6.3 in the Biochar Production and Storage Protocol	Independently verifi LCAs for the materia product completed; environmental prod declaration (EPD) fo material or product completed and independently verifi
Reactor temperature	Sensors for the monitoring or reactor temperature	Required	-	Measured	°C	Section 9.1.1 of the Biochar Production and Storage Protocol	Direct temperature measurements In lin temperature sensor
Pressure	Pressure measurements of the reactor	Required	-	Measured	Bar, Pa	Section 9.1.1 of the Biochar Production and Storage Protocol	Direct pressure measurements In lin pressure gauge or n
Gas flow	Gas flowmeters	Required	-	Measured	m^3 hr^{-1}	Section 9.1.2 of the Biochar Production and	Direct flow measurements Emis

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
						Storage Protocol	testing results Mate balance calculation
SOIL BASELINING							
Soil pH	Level of acidity or alkalinity of soil	Recommended	-	Measured	1 - 14	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Measurement of aci or alkalinity of soil
Soil moisture content	Moisture content of soil that biochar will be applied	Recommended	-	Measured	wt. %	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Analytical measurer of soil moisture con
Bulk density	Level of soil compaction and pore space availability	Recommended	-	Measured	g cm^{-3} or kg m^{-3}	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Analytical measurer of bulk density of th

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
Soil type and texture	Type of soil quality (sand, silt, clay) and the size distribution	Recommended	-	Measured	-	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Analytical measurer of soil type and text
Nutrient availability	Concentration of essential plant nutrients in forms that plants can readily absorb	Recommended	-	Measured	-	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Analytical measurer of nutrients in bioch available to plant fo absorption

PARAMETER	DESCRIPTION	REQUIRED OR RECOMMENDED	THRESHOLDS	PARAMETER TYPE	UNITS	PROTOCOL OR MODULE REFERENCE	DATA SOURCE
Soil Organic Carbon (SOC)	Indicator of soil organic matter and overall soil health	Recommended	-	Measured	g kg^{-1} or ppm	Section 5.2.2 of the Biochar Storage in Soil Environments Module	Analytical measurer of total organic carb content in soil
Crop yield or farm profitability data for three years proceeding biochar application	Time series of crop yield data	Recommended	-	Measured	Various accepted	Section 5.2.2 of the Biochar Storage in Soil Environments	Records of crop yie data

9.0

Appendix II: Guidance on Evidence for Biochar Application or Mixing

9.1

Introduction

This Appendix is a self-contained document that sets out comprehensive requirements for the types and quality of evidence necessary for projects to demonstrate proof of end use, either through application directly to soils, or incorporation of biochar into organic amendments (e.g., compost). Its primary objective is to ensure the durability and credibility of carbon storage claims made by biochar projects. At the point of mixing with either soil or organic amendments, the risk of reversal of total reversal is considered to be minimal. Adherence to these requirements is essential for establishing rigorous monitoring, reporting, and verification (MRV) systems that uphold transparency and trust within carbon markets and related environmental initiatives. These guidelines apply to all projects seeking to quantify and claim carbon Credits arising from the application or mixing of biochar in soils and must be adhered to.

Project Proponents must confirm which mixing pathway(s) will be used in their PDDs.

Project Proponents should provide sufficient evidence that the biochar is destined for application to soil, in addition the suggested evidence contained within this Appendix. If biochar or biochar-containing products are transferred from the Production Facility through intermediaries (i.e., entities other than the end user), monitoring of product must be maintained until delivery to the final user (and place for storage). The Project Proponent should establish procedures and agreements with all intermediaries to ensure the collection of complete traceability data.

In cases where intermediaries engage in additional processing activities, such as manufacturing new biochar products, blending biochar from multiple sources, or modifying its properties (e.g., via chemical or thermal treatment), these actions must be explicitly documented and reported to The Project Proponent.

9.2

Crediting at the Point of Mixing and Transfer

9.2.1

Principle

Credits will be issued at the point where biochar has been physically and irreversibly integrated into a final product, and legal custody of that product has been transferred to a third party. This is permitted because the mixing process, when performed according to the

requirements below, serves as a sufficient mitigation measure against the primary reversal risk of diversion for use as fuel.

The act of mixing and transferring the product is considered the point of durable storage, as the biochar is no longer in a pure, easily diverted form and the risk of reversal is minimized. There are three pathways to durable biochar storage in soils, direct spreading of the unamended biochar product, mixing with organic materials, or selling to a third party for use in agronomic products, the evidence requirements for which are specified in Sections 9.5, 9.6 and 9.7, respectively.

For application direct to soil Project Proponents must provide offtakers or downstream receivers with a best practice guidance for handling and application to ensure the risk of harm to the environment and human health is minimized. For projects mixing with organic materials, or selling to a third party for use in agronomic products it is highly recommended that this documentation is also provided.

9.2.2
Requirements for Crediting at Point of Mixing

For a project to claim Credits at the point of mixing and transfer, The Project Proponent must meet all of the following conditions for each batch of biochar:

- Irreversible product integration: The biochar must be homogenously mixed into a final product where it is no longer practically or economically feasible to separate it for alternative uses.
  - The biochar content must constitute less than 50% of the final product by volume.
  - The final product (e.g., compost, soil blend) must be unsuitable for use as a fuel due to its composition and moisture content.

- Defined end-Use pathway: The final mixed product must have a single, clearly defined end use that is an eligible storage pathway under this Module (e.g., agricultural soil amendment, landscaping material).

- Verified transfer of custody: The Project must demonstrate that the final mixed product has been sold or legally transferred to a third-party entity (e.g., a composting facility, agricultural distributor, or end-user) and that there is an clear burden of evidence that the biochar will be applied to a soil environment.

- Controlled end-of-life use: The once the biochar is incorporated into the final product there must be a burden of proof that the material will be applied to land and not disposed of through pathways that will result in reversal (e.g., municipal waste incineration).

- This pathway is not permitted in jurisdictions where waste incineration or energy-from-waste is the predominant disposal route, unless The Project Proponent can provide evidence that the biochar is ultimately incorporated into soil.

9.3 Definitions

- MRV: The multi-step process to monitor the Removals and impacts of a Project, report the findings to an accredited third party Validation and Verification Body (VVB), and have this VVB Verify the report so that the results can be Certified.
- Biochar Application: The direct deployment and incorporation of biochar into soil, typically for agricultural, land or environmental improvement purposes, where the biochar remains within the soil matrix.
- Mixing (with compost or other organic mixtures): The process of combining biochar with organic materials (e.g., compost, manures, biosolids) to create a blended product that is subsequently applied to soil. The biochar is integrated into the organic matrix prior to final land application.
- Project Boundaries: The clearly defined geographical areas (e.g., specific fields, farms, or land parcels) where biochar application or mixed product application activities occur.
- Third-Party Transactions: The sale or transfer of biochar from The Project developer or producer to an entity (the "Purchaser") that is not directly part of The Project's operational structure, and where the purchaser is responsible for the final application or mixing.
- Geotagged Evidence: Digital media (e.g., photos, videos) that include embedded metadata providing precise geographical coordinates (latitude, longitude) and a timestamp, confirming the location and time of capture.

9.4 General Principles for Evidence

All evidence submitted for biochar application or mixing projects must adhere to the following general principles to ensure accuracy, traceability, and verifiability:

- Traceability and Transparency:
  - All data and documentation must clearly link the biochar from its production to its final application or mixing point. This must be done at the storage batch level, but should be traceable to the production batch.
  - The entire chain of custody must be transparent and auditable.
- Timeliness and Consistency:
  - Data must be collected at the time of each biochar application or mixing event. Evidence must be collected and presented in a

manner consistent with Isometric's policy for data collection and storage outlined in the Biochar Production and Storage Protocol and the Biochar Storage in Soil Module.
    - All records must be maintained for a minimum of 5 years from the date of collection, using standardized formats to ensure completeness, comparability, and reliability over time.
- Acceptable Formats and Quality Standards:
    - Evidence should be clear, legible, and unambiguous.
    - Digital formats are preferred, and physical records must be scanned and stored digitally.
    - Photos and videos must be of sufficient resolution and clarity to demonstrate the activity.

9.5 Acceptable Evidence for Biochar Application Directly to Soil

Project Proponents must document either of the following methods detailed in Section 9.5.1 or 9.5.2 that are accepted as sufficient evidence of spreading of biochar:

9.5.1 Visual Documentation

Geotagged photos or videos are critical for visually confirming the application of biochar. These must include all of the following for every storage batch:

- Biochar Stockpiles Before Application:
    - Images showing the biochar material (e.g., in bags, piles, or storage containers) at the application site, clearly identifiable as biochar, prior to its spreading.
- Biochar Being Spread or Mixed:
    - Visuals capturing the active process of biochar being applied to the land (e.g., by spreader, tractor, or manual labor) or being incorporated into the soil.
- Final Incorporation into Soil or Organic Matrix:
    - Photos or videos demonstrating the biochar after it has been fully incorporated into the soil or mixed into an organic matrix, showing the uniformity of application.
- Requirements for Geolocation Metadata and Time Stamps:
    - All visual documentation must have embedded GPS coordinates (latitude, longitude) and accurate time and date stamps.
    - This metadata must be verifiable and consistent with the project boundaries and application records. If metadata is not automatically embedded, a separate log linking image filenames to GPS coordinates and timestamps must be maintained.

9.5.2

Project Boundary Mapping With Application Records

Project boundaries are required to define the areas of application:

- Maps or Geographic Information System (GIS) Layers
  - High-resolution maps or GIS layers clearly delineating the area of land where biochar application occurred. If the landowner requests anonymity, proof can be provided at the ZIP code (or equivalent) level.
  - These maps should include relevant identifiers (e.g., field names/numbers, land parcel IDs).
  - Spreading is not permitted outside of the project boundaries agreed in the PDD.

And, complete logbooks or digital databases are required to detail application events:

- Dates and Quantities Applied:
  - Dates of application and the quantity of biochar (e.g., in tonnes) applied to each specific area evidenced by weighbridge or inventory records or affidavit, from which an application rate can be calculated.
  - N.B. The application rate should not exceed the maximum loading rates established by the relevant jurisdiction where the biochar is being applied.

9.6

Acceptable Evidence for Mixing With Organic Material (e.g., Compost)

When biochar is mixed with other organic materials the biochar content must be below 50% (v/v) of the final product and shall ensure that the intermixture cannot self-sustain combustion. Project Proponents must document all of the following methods detailed in Section 9.6.1 and 9.6.2, unless listed as optional:

9.6.1

Mixing Facility Documentation

If separate to the production facility:

- Records of the Facility:
- Documentation confirming the location, operational license (if applicable), and capacity of the facility where the biochar and organic materials are mixed.
- Batch Records:
  - Detailed records linking specific batches of incoming biochar (with unique storage batch IDs) to ensure traceability of the

biochar.

9.6.2
Mixing Process Evidence

- Photos/Videos of Biochar Integration:
  - Visual evidence showing the biochar being actively integrated into the organic mixture (e.g., during windrow turning, mechanical blending) to be provided at least every storage batch. Isometric also reserves the right to request further photo and video evidence on a random basis.
- Weighbridge or Inventory Records:
  - Records from weighbridges or other inventory management systems demonstrating the quantities of biochar and organic materials used as inputs, and the quantity of the final mixed product produced.
- Quality Control or Laboratory Records (Optional but Recommended):
  - Laboratory analyses of the final mixed product to confirm biochar content (e.g., through proximate analysis, ash content, or other suitable methods) compared to the unamended product.

9.7
Evidence When Biochar Is Sold to Third Parties

When raw biochar is sold or transferred to a third party for mixing with other organic material (e.g. compost) or other agronomic products, additional documentation is required to ensure accountability and traceability. The biochar content must be below 50% (v/v) of the final product and shall ensure that the intermixture cannot self-sustain combustion. Project Proponents must document all of the following methods detailed in Section 9.7.1 and 9.7.2:

9.7.1
Affidavits or Declarations

- Affidavit from Purchaser: A legally binding affidavit or declaration signed by the purchaser of the biochar, confirming:
  - The explicit intended use of the biochar (e.g., direct application to soils, mixing at a composting facility, or other specified carbon-sequestering use).
  - The maximum geographical boundaries or specific locations where the biochar will be applied or stored. This will be used to conservatively estimate the soil temperature if The Project Proponent plans to use the 200 year crediting option, using the highest soil temperature for the region or location.
  - An agreement to retain and provide supporting evidence of application or mixing if requested by The Project Proponent or VVB.

- As well as at least 3 years of evidence that the company receiving the biochar is an active agricultural or adjacent company. Or, alternative information to show the company has legitimate end use of the biochar.
- The biochar processing facility will mix the biochar with the material, to reduce the risk of reversal, within a stated number of days of receipt of the biochar. Credits will not be issued against this biochar until this time period during which mixing should occur has elapsed.

9.7.2
Sales Documentation

- Sales Invoices or Transfer Records:
  - Sales invoices, purchase orders, or transfer records that clearly link the volume and unique batch IDs of the biochar sold to the specific purchaser.
- Proof of Delivery: Documentation confirming the delivery of the biochar to the third party's specified location, such as:
  - Bills of lading (BOLs).
  - Delivery receipts signed by the recipient, or proof of delivery.
  - For bulk deliveries, GPS data from delivery vehicles or geotagged photos at the delivery site (if feasible).

9.7.3
Mixing Process Evidence

- Photos/Videos of Biochar Integration:
  - Visual evidence showing the biochar being actively integrated into the organic mixture (e.g., during windrow turning, mechanical blending) to be provided at least once per verification. Isometric also reserves the right to request further photo and video evidence on a random basis.
- Weighbridge or Inventory Records:
  - Records from weighbridges or other inventory management systems demonstrating the quantities of biochar and organic materials used as inputs, and the quantity of the final mixed product produced.
- Quality Control or Laboratory Records (Optional but Recommended):
  - Laboratory analyses of the final mixed product to confirm biochar content (e.g., through proximate analysis, ash content, or other suitable methods) compared to the unamended product.

9.8
Guidance on Transport Emissions (only Applicable to Biochar Mixed With Organic

Material, or Sold to Third Parties)

In line with Section 8.1 of the Biochar Storage in Soils Module and Section 7.1.1.4 of the Biochar Production and Storage Protocol, transport emissions from the processing facility (e.g. the composting plant) to the end use location must be included in the project boundary if:

- The average transport distance for biochar-amended compost is greater than the average transport distance for conventional products made at the same facility.

Exclude downstream transport emissions if:

- These activities were already happening before The Project.
- They would still happen even if The Project did not exist.

Evidence requirement:

- Provision of documentation showing that the biochar-amended product performs to at least the same standard as a conventional product for the same intended use.
- Evidence of the average biochar transport distance not being greater than the equivalent conventional products.

9.9

Chain-Of-Custody Requirements

Project Proponents must provide a clear chain of custody diagram or equivalent.

A robust chain-of-custody system is paramount for tracking biochar from production to final application. Implementation must include:

- Unique batch identification numbers:
  - Every batch of biochar produced must be assigned a unique, sequential identification number. This ID must be used on all related documentation.

- Documentation at each handoff:
  - Clear documentation is required at every point where custody of the biochar changes hands (e.g., From The Project Proponent to third-party purchaser). Examples of adequate evidence includes:
    - Dispatch notes
    - Receiving reports
    - Quality control certificates (if applicable)
    - Signed delivery notes or bills of lading or invoices

- Record retention period:
  - All chain-of-custody documentation, along with all other evidence, must be retained for a minimum period (5 years from the point of creation). Records should be stored securely and be readily accessible for audit.

9.10 Verification and Audit Guidance

VVBs will examine all submitted evidence to ensure compliance with the PDD, the Biochar Production and Storage Protocol and Biochar Storage in Soil Module.

- What VVBs Will Check:
  - Consistency between application records, visual evidence, and mapping
  - Accuracy of quantities reported against sales/delivery records
  - Completeness of chain-of-custody documentation
  - Verification of geotagging metadata
  - Interviewing personnel involved in biochar application or mixing
  - Site visits to confirm application areas and practices
- Common Evidence Gaps and How to Avoid Them:
  - Missing Geotags: Ensure all photos/videos are geotagged correctly. Use apps that automatically embed metadata or maintain a strict manual log.
  - Inconsistent Data: Implement standardized data collection forms and train personnel thoroughly to ensure consistency across all records.
  - Incomplete Chain-of-Custody: Document every transfer of biochar, no matter how small.
  - Lack of Specificity: Avoid vague descriptions. Be precise with dates, quantities, locations, and methods.
  - Poor Quality Visuals: Ensure photos and videos are clear, well-lit, and show the relevant activity.
- Requirements for Maintaining Evidence in Auditable Form: All evidence must be organized, indexed, and stored in a manner that facilitates easy retrieval and review by auditors. Digital storage with proper backup Protocols is highly recommended.

9.11 Optional Enhanced Evidence

Project developers may choose to provide additional evidence to further strengthen the credibility of their data:

- Remote Sensing or Drone Imagery:

    - High-resolution satellite or drone imagery can provide an independent means of verifying application areas and, in some cases, the presence of applied material, especially for large-scale projects.
- Laboratory Analysis of Soils or Compost for Biochar Content:
    - Independent laboratory analysis of soil samples from treated areas or samples of the biochar-amended compost to confirm the presence and quantity of biochar. While current methods cannot accurately separate soil organic matter and biochar organic matter, biochar application and incorporation will increases total organic matter/carbon, though this is dependent on the application rate.
- Blockchain or Digital Traceability Tools:
    - Utilizing digital traceablity technology or other secure digital platforms to record and verify biochar transactions and application events can provide an immutable and highly transparent chain of custody.

10.0

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11.0

Definitions and Acronyms

Activity
The steps of a Project Proponent's Removal or Reduction process that result in carbon fluxes. The carbon flux associated with an activity is a component of the Project Proponent's Protocol.

American Society for Testing and Materials (ASTM)
A standards organization that develops and publishes voluntary consensus international standards.

Buffer Pool
A common and recognized insurance mechanism among Registries allowing Credits to be set aside (in this case by Isometric) to compensate for Reversals which may occur in the future.

Carbon Dioxide Equivalent Emissions (CO_2e)
The amount of CO_2 emissions that would cause the same integrated radiative forcing or temperature change, over a given time horizon, as an emitted amount of GHG or a mixture of GHGs. One common metric of CO_2e is the 100-year Global Warming Potential.

Carbon Dioxide Removal (CDR)
Activities that remove carbon dioxide (CO_2) from the atmosphere and store it in products or geological, terrestrial, and oceanic Reservoirs. CDR includes the enhancement of biological or geochemical sinks and direct air capture (DAC) and storage, but excludes natural CO_2 uptake not directly caused by human intervention.

Carbon Flux
The amount of carbon exchanged between two or more Reservoirs over a period of time.

Cation Exchange Capacity (CEC)
A measure of a soil's ability to hold and exchange cations.

Conservative
Purposefully erring on the side of caution under conditions of Uncertainty by choosing input parameter values that will result in a lower net CO_2 Removal or GHG Reduction than if using the median input values. This is done to increase the likelihood that a given Removal or Reduction calculation is an underestimation rather than an overestimation.

Credit
A publicly visible uniquely identifiable Credit Certificate Issued by a Registry that gives the owner of the Credit the right to account for one net metric tonne of Verified CO_2e Removal or Reduction. In the case of this Standard, the net tonne of CO_2e Removal or Reduction comes from a Project Validated against a Certified Protocol.

Crediting Period
The period of time over which a Project Design Document is valid, and over which Removals or Reductions may be Verified, resulting in Issued Credits.

Double Counting
Improperly allocating the same Removal or Reduction from a Project Proponent more than once to multiple Buyers.

Durability

The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question.

Emissions
The term used to describe greenhouse gas emissions to the atmosphere as a result of Project activities.

Feedstock
Raw material which is used for CO_2 Removal or GHG Reduction.

GHG Statement
A document submitted alongside Claimed Removals and/or Reductions that details the calculations associated with a Removal or Reduction, including the Project's emissions, Removals, Reductions and Leakages, presented together in net metric tonnes of CO_2e per Removal or Reduction.

Global Positioning System (GPS)
A satellite-based navigation system.

Greenhouse Gas (GHG)
Those gaseous constituents of the atmosphere, both natural and anthropogenic (human-caused), that absorb and emit radiation at specific wavelengths within the spectrum of terrestrial radiation emitted by the Earth's surface, by the atmosphere itself, and by clouds. This property causes the greenhouse effect, whereby heat is trapped in Earth's atmosphere (CDR Primer, 2022).

ICP-MS
Inductively Coupled Plasma Mass Spectrometry: An analytical technique used to measure elements at trace levels within a sample.

International Standards Organization (ISO)
A worldwide federation (NGO) of national standards bodies from more than 160 countries, one from each member country.

Issuance (of a Credit)
Credits are issued to the Credit Account of a Project Proponent with whom Isometric has a Validated Protocol after an Order for Verification and Credit Issuance services from a Buyer and once a Verified Removal or Reduction has taken place.

Losses
for open systems, biogeochemical and/or physical interactions which occur during the removal process that decrease the CO_2 removal .

Model
A calculation, series of calculations or simulations that use input variables in order to generate values for variables of interest that are not directly measured.

Module
Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Monitoring, Reporting, and Verification (MRV)
The multi-step process to monitor the Removals or Reductions and impacts of a Project, report the findings to an accredited third party, and have this third party Verify the report so that the results can be Certified.

Pathway
A collection of Removal or Reduction processes that have mechanisms in common.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Design Document
The document, written by a Project Proponent, which records key characteristics of a Project and which forms the basis for Project Validation and evaluation in accordance with the relevant Certified Protocol. (Also known as “PDD”).

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Project boundary
The defined temporal and geographical boundary of a Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Proxy
A measurement which correlates with but is not a direct measurement of the variable of interest.

RP
Reporting Period

Remote Sensing
The use of satellite, aircraft and terrestrial deployed sensors to detect and measure characteristics of the Earth's surface, as well as the spectral, spatial and temporal analysis of this data to estimate biomass and biomass change.

Removal
The term used to represent the CO_2 taken out of the atmosphere as a result of a CDR process.

Reversal
The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued. A Reversal is classified as avoidable if a Project Proponent has influence or control over it and it likely could have been averted through application of reasonable risk mitigation measures. Any other Reversals will be classified as unavoidable.

SOC
Soil Organic Carbon

Sink
Any process, activity, or mechanism that removes a greenhouse gas, a precursor to a greenhouse gas, or an aerosol from the atmosphere.

Source
Any process or activity that releases a greenhouse gas, an aerosol, or a precursor of a greenhouse gas into the atmosphere.

Storage

Describes the addition of carbon dioxide removed from the atmosphere to a reservoir, which serves as its ultimate destination. This is also referred to as “sequestration”.

System Boundary
GHG sources, sinks and reservoirs (SSRs) associated with the project boundary and included in the GHG Statement.

Uncertainty
A lack of knowledge of the exact amount of CO_2 removed by a particular process, Uncertainty may be quantified using probability distributions, confidence intervals, or variance estimates.

Validation
A systematic and independent process for evaluating the reasonableness of the assumptions, limitations and methods that support a Project and assessing whether the Project conforms to the criteria set forth in the Isometric Standard and the Protocol by which the Project is governed. Validation must be completed by an Isometric approved third-party (VVB).

Validation and Verification Bodies (VVBs)
Third-party auditing organizations that are experts in their sector and used to determine if a project conforms to the rules, regulations, and standards set out by a governing body. A VVB must be approved by Isometric prior to conducting validation and verification.

Verification
A process for evaluating and confirming the net Removals and Reductions for a Project, using data and information collected from the Project and assessing conformity with the criteria set forth in the Isometric Standard and the Protocol by which it is governed. Verification must be completed by an Isometric approved third-party (VVB).

12.0

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v1.2

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This module is an expansion of the requirements from Biochar Storage in Agricultural Soils, and hence starts on version 1.2. Key requirement changes from Biochar Storage in Agricultural Soils version 1.1 include scope expanding beyond agricultural soils to include forestry soils and recreational areas; new provisions for stockpiling, mixing with organic materials, and supply chain sales; updates to carbon fraction calculations and measurement guidance. Structural change, module title updated to Biochar Storage in Soil Environments Module. Text formatting updates

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